WP3: LLM Provisioning

Feasibility of state-of-the-art and alternative model options for SWISSGEO conversational access

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Legends in this report

Table 1. Verdict wording used in tables

Verdict wording used in tables	What it means
Ready	Production-ready, or the recommended choice
Caveat	Usable, but with a stated limitation, or worth watching
Not viable	Ruled out for this project specifically
Reference	Useful for comparison; not a deployment candidate

Table 2. Jurisdiction tag

Jurisdiction tag	Meaning
[CH]	Swiss jurisdiction (headquartered, trained, or hostable in Switzerland)
[EU]	European Union jurisdiction
[US]	United States jurisdiction
[CN]	Chinese jurisdiction

This tag is a simplification of something more layered. Who legally owns the vendor, where processing physically happens, and where the model was trained aren't always the same country. §6 unpacks all three properly, with the legal entity behind every model in this report and what it implies.

Every model and platform table also carries an **openness tier**:

Table 3. Openness tier

Openness tier	What it means	Example
Open-source	Weights, training data *and* code are all released	Apertus, OLMo, EuroLLM
Open-weight	Only the trained weights are released; training data and code are not	Mistral, gpt-oss, DeepSeek, Qwen, GLM, Kimi
Closed	Available only as a paid API; nothing is released	Claude, GPT, Gemini, Grok, Nova
Restricted	Weights released, but under a license that limits who may use them	Llama 4

Cost figures follow rule: **Low / Central / High**, where Low and High are floor-and-ceiling estimates - the cheapest and most expensive realistic setups we could find, not a narrow band around the middle. A single-point figure is called out as such rather than dressed up as a range.

Contents

1. Summary

This report answers one question for swisstopo: which LLM (the AI system that understands and answers a user's question) should power SWISSGEO's conversational access to the geo.admin.ch catalogue, and what are the credible alternatives.

> **Recommendation:** deploy Anthropic's Claude models via Amazon Bedrock, Amazon's managed service for running AI models and already part of swisstopo's infrastructure, using its EU-resident routing option. Bedrock calls these "`eu.` profiles": the request is guaranteed to process inside EU data centres rather than route globally. Start with the mid-tier model, Sonnet 4.6, as the default. Keep a watch on European, Chinese-origin and Swiss (Apertus) alternatives, and potentially switch if the situation changes.

Cost. At a realistic production volume, Sonnet 4.6 runs roughly CHF 690–1,030/month non-optimized, potentially falling to CHF 180–270/month when the optimization stack is built; see §4.3 for the full derivation, §4.5 for the optimization stack, and §4.7 for the higher-volume case.

Answer quality on multi-step tasks. Where models actually differ is in chaining several tool calls together, recovering from an error mid-conversation, and knowing when to say "no data exists" instead of guessing; agentic readiness is best reviewed in §5.7 (Table 25).

Sovereignty. It has many steps from a cloud API to fully on-premises hardware, and each step trades cost against control differently. Our recommendation is the first step (§6.1, Table 26).

Table 4. Summary questions and answers

Question	Answer	Confidence
What should we deploy?	Claude Sonnet 4.6 via Amazon Bedrock, EU-resident routing, on swisstopo's existing AWS account	High, closest to contract requirement; matches a live government precedent (GOV.UK Chat)
What will it cost?	Central estimate is CHF 690–1,030/month non-optimized, potentially CHF 180–270/month optimized, at a realistic production volume (full breakdown in §4.3, optimization stages in §4.5, higher-volume case in §4.7, pilot cost in §4.3's table)	High for the unit-cost arithmetic; Medium for adoption volume and query complexity, both our best estimate until pilot data exists; the optimized figures depend on which get built and their effectiveness
How sovereign?	It's the first step of five. Every step uses the same API, so moving between them is a configuration change after setup (§6.1, Table 26)	High
What's the fallback?	Mistral Large 3 (European, open-weight) is the strongest alternative on current evidence, but it's a clear step down from Claude on demonstrated agentic capability, not a peer contender; Apertus (Swiss, open-source) not yet viable (§5.3 Table 20; §5.7 Table 25)	High on the ranking; Medium on timing, since it depends on Apertus v1.5, which has no release date and unknown performance

2. Scope

WP3 of the Phase-2 contract asks for two things: a state-of-the-art LLM provisioned on swisstopo's own AWS infrastructure, and a consideration of the feasibility of alternatives, naming open-source and open-weight models such as OLMo, and the Swiss model Apertus, explicitly.

It also answers three other pieces of client input directly:

- **SGS-5-25-2** (24.11.2025), swisstopo's own decision-basis report for integrating LLMs into the national geodata infrastructure, which sets the governance constraints this report works within (§3, Table 6).
- **Five questions**: Bedrock cost, open-model cost, whether a large model is even needed, whether a small self-hosted model would suffice, and how to control cost; answered across §4 and §5.

2.1 Document map

Table 5. Document map

Source document	What it asked for	Where this report answers it
Phase-2 contract, WP3 clause	SotA LLM deployment plus an alternatives report (incl. OLMo, Apertus)	Whole document; §5.3–§5.7 for the alternatives
SGS-5-25-2	Governance constraints, risk classification, sovereignty preference	§3 (Table 6), §6.1–§6.2
Five coordination questions (06.07.2026)	Bedrock cost / open-model cost / model size / self-hosting / cost control	§4.3, §4.5, §4.8, §5.2–§5.7

3. Requirements and constraints

Table 6. Requirements and constraints

Constraint	Source	Binding or preferred?	Addressed in
Deployed on swisstopo's AWS infrastructure; open-source repository on swisstopo's GitHub	Contract	Binding	§6.1 (Table 26, step 1 — Bedrock `eu.` profiles); §7.1 (Table 30, Level 1)
Only swisstopo data: no OpenStreetMap, no fusion with outside sources; the assistant adds value to the existing geoadmin API rather than replacing it	PoC kickoff	Binding	§2 (scope boundary) — a tool/API-design constraint, outside this report's model-comparison tables
Every answer cites its source; the assistant says plainly when no matching data exists rather than guessing	PoC kickoff, SGS report	Binding	§5.1 (refusal/error-recovery benchmark evidence [23][24])
Coordinate system EPSG:2056 (the Swiss national grid); catalogue of roughly 16,000 datasets / 800+ layers	PoC kickoff	Binding	§5.1 (geospatial tool-use and long-context benchmark evidence [20][25]) — context for scale, not a dedicated table
Working languages: German, French, Italian, English. Romansh and Swiss German are Swiss-specific ambitions, keep updated	PoC kickoff; WP5 test plan	Binding (DE/FR/IT/EN); keep updated (RM/gsw)	§5.6 (Table 24, Romansh/Swiss German rows); §7.1 (Table 30, Level 2 gate — WP5 evaluation)
Risk classification: limited risk, since the data handled is open, non-personal, and the assistant is read-only and informational	SGS-5-25-2	Shapes governance weight	§6.2 (public, non-personal data framing)
Sovereignty: data handling, model training and operational use should occur as far as possible on national or European infrastructure	SGS-5-25-2	Preferred	§6.1 (Table 26, the sovereignty scale)
eCH-0272, the Swiss standard for transparency in AI systems, applies to pre-trained, locally-operated systems - cloud-only GenAI is outside its scope	eCH-0272 standard	Contextual	§6.2 (provenance and standards discussion)
Test-case-driven decisions: the SGS report's own mechanism for choosing a model is a dedicated evaluation, not a specification on paper	SGS-5-25-2	Binding on method	§7.1 (Table 30, WP5 evaluation gate)

4. Cost

4.1 Three ways to pay for an LLM

Table 7. Three ways to pay for an LLM

Mode	What it is	Who owns the infrastructure	What you're billed for	Minimum realistic commitment	When it's the right choice
Serverless API (Amazon Bedrock, or a vendor's own API)	You send a request to a model someone else runs; you never touch a server	The cloud vendor	Per token	None - pay for what you use	Any volume, especially traffic that's uncertain or bursty - **this project's situation**
Self-hosted server (Amazon SageMaker, or a rented cloud GPU)	You rent a GPU-equipped machine and run the model's software on it yourself	You, on the vendor's hardware	Per hour the machine is switched on, regardless of how much it's used	The machine typically needs to run near-continuously to be worth deploying	An open-weight model isn't available serverless or inference has to be in a specific country
On-premises hardware	You buy physical GPU servers and run them in your own building or a data centre you control	You, entirely	Hardware capex, power, and salaries	A multi-year hardware purchase and a standing operations team	Only when physical, air-gapped control is mandated (§4.8)

4.2 How many conversations will there be?

The public `sgs-llm` cost model scales "40–80,000 portal hits/day" into scenarios of up to 20,000 assistant conversations per day. Published data from comparable, already-live government assistants suggests that's about two orders of magnitude too high:

Table 8. Comparable government-assistant deployments

Deployment	Scale	Published volume	Rate
GOV.UK Chat (Claude on Bedrock) [5]	National (68 million population)	15,000+ questions, 7-week soft launch	~306 questions/day
ZüriCityGPT (Liip, City of Zurich) [7]	City-level	39,000 conversations in 8 months	~160 conversations/day
RNP assistant, City of Riga [69]	Municipal (housing authority), not national	~47,000 conversations in 5 months	~310 conversations/day
GOV.UK, 18-month pilot phase [6]	National	26,000 questions, 10,000+ users	Accuracy bar rose 76% → 90% over the pilot

This report grounds Low and Central directly in what the comparable deployments above actually recorded, then uses swisstopo's own real traffic, not a generic industry benchmark, to set a High that reflects swisstopo's genuinely larger reach, rather than either ignoring that traffic entirely or projecting it through an unrelated benchmark:

Table 9. Low / Central / High volume assumptions

	Low	Central	High
Basis	ZüriCityGPT [7], the smallest and only other Swiss deployment in this set	Simple average across the three live deployments with a daily rate (ZüriCityGPT, GOV.UK Chat, RNP Riga) [5][7][69]	Swisstopo's own traffic [4], at the same conservative, task-driven open rate that independently reproduces Central (see below)
Conversations/ day	~160	~260	~452 (90,400 visits/day × 0.5% open rate)
Turns/day	**~440** (2.75 turns/conversation)	**~715** (2.75 turns/conversation)	**~1,350** (Larger, more habitual audience plausibly runs longer sessions)

Low and Central are narrow (roughly 2× apart) because that's what the evidence actually shows: GOV.UK Chat (national, 68 million population) and RNP Riga (a single municipal housing authority) land within a few percent of each other despite a huge difference in the population each one serves, and ZüriCityGPT isn't far below either. Reach doesn't appear to scale with population for this category of assistant

High: map.geo.admin.ch recorded 33 million visits in 2025, ~90,400/day [4]. A generic live-chat benchmark (2–8%, up to 15% with pop-ups [58]) would imply several thousand turns/day (far above any comparable government assistant on record), so this report instead uses a conservative, portal-adjusted open rate of 0.5%, reflecting that geodata-portal visitors mostly arrive already knowing what they want. That rate cross-checks against precedent: a similarly conservative ~0.3% rate, applied with the midpoint turns/conversation to swisstopo's own traffic, independently reproduces ~680 turns/day, within 5% of Central, two unrelated methods converging on the same place. High simply carries that logic to the top of the plausible range (0.5% open rate, 3 turns/conversation) on swisstopo's real traffic.

4.3 What does one conversation cost, before optimization?

Pricing basis: a typical agent turn (one question, including its internal tool calls) uses about 5,000 input tokens and 300 output tokens; a complex, multi-step turn uses 2–3× that. Vendor list prices are quoted in USD (Anthropic's own currency for Bedrock), then converted to CHF for every cost estimate in this report at the current USD/CHF rate (≈0.81, pulled 14.07.2026 [73]) to avoid mixing currencies. (§4.6's price-history table below is the one exception, left in USD since it quotes vendors' own published list prices as a directly checkable historical record.) Figures are per Claude model, on Bedrock, EU-resident routing (+10% over the global rate), pulled 14.07.2026 [9]. For example, Sonnet 4.6 at $3/$15 per million tokens (≈CHF 2.43/CHF 12.15) costs 5,000 × CHF 2.43/1,000,000 + 300 × CHF 12.15/1,000,000 ≈ CHF 0.0158 per lean turn; at Central (715 turns/day × 30.4 days/month, the 365/12 average used consistently throughout this report) that's 21,748 turns/month × CHF 0.0158 ≈ CHF 344/month. Monthly figures throughout this report use that same 30.4 days/month, so monthly × 12 always reconciles with the annual figures quoted elsewhere:

Table 10. Per-model pricing, naive (no caching, no routing)

Model	CHF per 1M input / output tokens	CHF/turn	Pilot (~60 turns/day)	Low (440 turns/day)	Central (715 turns/day)	High (1,350 turns/day)
Haiku 4.5 (the potential "routing tier")	0.81 / 4.05	0.0053	~CHF 10/mo	~CHF 70/mo	~CHF 114/mo	~CHF 216/mo
Sonnet 4.6 (simple query average)	2.43 / 12.15	0.0158	~CHF 29/mo	~CHF 211/ mo	~CHF 344/mo	~CHF 649/mo
Sonnet 4.6, complex, multi-step queries (2–3× tokens/turn)	2.43 / 12.15	0.0316–0.0474	~CHF 58–86/ mo	**~CHF 423–634/mo**	**~CHF 687–1,030/mo**	**~CHF 1,297–1,946/mo**
Sonnet 5 (intro price to 31.08.2026, then CHF 2.43/12.15)	1.62 / 8.10	0.0105	~CHF 19/mo	~CHF 141/ mo	~CHF 229/mo	~CHF 432/mo

Model	CHF per 1M input / output tokens	CHF/turn	Pilot (~60 turns/day)	Low (440 turns/day)	Central (715 turns/day)	High (1,350 turns/day)
Opus 4.8 (quality ceiling probe)	4.05 / 20.25	0.0263	~CHF 48/mo	~CHF 352/mo	~CHF 573/mo	~CHF 1,081/mo
Mistral Large 3 (La Plateforme, EU vendor, simple query average) [74]	1.62 / 4.86	0.0096	~CHF 17/mo	~CHF 128/mo	~CHF 208/mo	~CHF 392/mo
Mistral Large 3, complex, multi-step queries (2–3× tokens/turn) [74]	1.62 / 4.86	0.0191–0.0287	~CHF 35–52/mo	~CHF 256–384/mo	~CHF 416–623/mo	~CHF 785–1,177/mo

Every figure above is naive: no caching, no routing, nothing.

4.4 The same workload, three ways to run it

Table 11. Deployment mode comparison, annual

Deployment mode	Low (CHF/year)	Central (CHF/year)	High (CHF/year)	Notes
Serverless (Bedrock, Sonnet 4.6, complex queries, naive)	~5,076–7,608	~8,244–12,360	~15,564–23,352	The simpler lean-question floor is lower, at ~2,532/4,128/7,788/year
Serverless (Bedrock, Sonnet 4.6, complex queries, optimized, previewing §4.5)	~1,320–1,980	~2,148–3,216	~4,044–6,072	Lean floor optimized: ~55–169/month
Serverless (Mistral Large 3, La Plateforme, complex queries, naive) [74]	~3,072–4,608	~4,992–7,476	~9,420–14,124	EU vendor, not swisstopo's existing AWS contract; not reachable via Bedrock in an EU region (§5.3 Table 19, §6.1 Table 26). Lean floor: ~1,536/2,496/4,704/year
Self-hosted server (SageMaker, open-weight model, 24/7 in Zurich)	~150,000	~205,000	~260,000	Fixed regardless of how many conversations actually happen
On-premises (small rig, business-hours operation)	~137,000	~208,000	~280,000	Fixed, three-year hardware amortization, 0.5–1 FTE of staff time
On-premises (production-scale, 8×H100-class rig)	~470,000	~625,000	~780,000	Only when requirement for physical control (§4.8)

The pattern holds at every volume this project is likely to see: serverless cost scales with actual use, while self-hosted and on-premises costs are fixed regardless of use, and at this precedent-grounded volume serverless remains two to three orders of magnitude cheaper even before optimization.

4.5 What brings the naive number down

Everything that follows is what can bring the naive estimate down with extra work. The numbers below reflect probable estimates, rather than a best case.

This project's recommended path is serverless (§4.4 above). The levers above the line apply to that path; the levers below it only matter for a fixed-cost self-hosted or on-prem deployment, which isn't being recommended here:

Table 12. Optimization levers

Lever	Mechanism	Realistic saving	Applies to
Prompt caching	Repeated system instructions and tool definitions are billed at a tenth of the price on re-use within a session	30–45% off the cost of calls that still reach a model	Serverless, self-hosted
Model routing (what some vendors market separately as "cascading" is the same mechanism: send easy questions to a cheap model, escalate the hard ones.) [13]	Simple questions answered by Haiku; hard ones escalate to Sonnet or Opus	Depends entirely on how much real traffic turns out to be simple, which is unproven for swisstopo's query mix until the evaluation and pilot logs exist. Published extremes (10–20× cheaper) assume most traffic is simple; this report uses a more modest working assumption until it's measured	All modes
Semantic answer caching [15]	A repeated question is answered from a stored prior answer, without calling a model at all	20–35% of turns removed, toward the conservative end of the 30–70% published range, since swisstopo-specific repeat-question rates aren't known yet	All modes
— not applicable to this project's recommended serverless path —			
Scale-to-zero scheduling	Infrastructure not mostly needed overnight or on weekends isn't paid for then	Relevant only where the bill is otherwise fixed	Self-hosted, on-prem
Reserved/committed pricing	Committing to a usage volume or instance-hours in advance	~60% off list, on infrastructure that's already rented long-term anyway	Self-hosted, on-prem

Table 13. Running total after each optimization stage

Stage	What optimizations applied	Running total (central est, Sonnet 4.6)
Naive	Nothing yet	CHF 687–1,030/month
+ Semantic caching	~25% of turns answered from cache, no model call	CHF 515–773/month
+ Prompt caching	~35% off whatever still reaches a model	CHF 335–502/month
+ Model routing	Of the remaining calls, roughly two-thirds go to Haiku at ~30% of Sonnet's cost	CHF 179–268/month

That's a reduction to roughly a quarter of the naive figure (factor 0.26, ÷3.8). Overlap between semantic and prompt caching is accounted for and routing is calibrated on real, not assumed, traffic. Semantic and prompt caching ship first since both are available directly from the cloud provider; model routing ships last since it requires building and tuning the routing logic itself. If only caching ships near-term, a more conservative interim figure is **CHF 335–502/month**, with routing added later once real traffic data justifies it.

Self-hosted server (SageMaker or similar). These levers barely move this bill: it's a rental, not a metered charge, so caching and routing only affect efficiency, not cost. The lever that matters here is scale-to-zero scheduling: running only during real usage hours instead of 24/7. Bedrock's own Custom Model Import pricing for a comparable model, active ~12h/day, gives ~CHF 95,000–110,000/year, against ~CHF 205,000/year continuous.

On-premises. No meaningful reduction applies: hardware and staff are paid for regardless of volume. Software efficiency only buys headroom (delaying the next hardware purchase or hire), not a lower standing bill.

This is why serverless is the default recommendation here: it's the only mode where efficiency gains lower the bill rather than just freeing up already-paid-for capacity.

4.6 What changes over time: capability, not the price tag

The industry habit is to project a single "AI inference gets X% cheaper per year" curve forward and apply it to today's bill. That curve is real at the level of compute economics, but it doesn't describe what happens to a specific, persisting deployment, for two reasons worth separating: the model itself doesn't sit still and get a discount (it gets retired), and the number of tokens a task actually consumes is moving too, mostly upward.

Models don't get cheaper: they get retired. Anthropic does not keep a fixed model on sale at a falling price; it publishes a retirement date, typically 12–18 months after release, and requires migration to whatever replaces it [70]:

Table 14. Claude model deprecation schedule

Model	Deprecated (notice given)	Retired (requests fail)
Claude Sonnet 3.5 (both releases)	13 Aug 2025	28 Oct 2025
Claude Sonnet 3.7	28 Oct 2025	19 Feb 2026
Claude Haiku 3.5	19 Dec 2025	19 Feb 2026
Claude Haiku 3	19 Feb 2026	20 Apr 2026
Claude Sonnet 4 / Claude Opus 4	14 Apr 2026	15 Jun 2026
Claude Opus 4.1	5 Jun 2026	5 Aug 2026 (scheduled)

On any planning horizon past roughly 18 months, Sonnet 4.6 (the model recommended in this report) will not be an option any more. swisstopo will be migrating to whatever the then-current Sonnet-tier model is and re-validating it against the WP5 test cases before switching over, regardless of whether the replacement turns out cheaper, the same price, or more expensive. Anthropic frames the retirements as a byproduct of keeping the frontier moving, not as a service guarantee, and has committed only to preserving retired models' weights for research, not to keeping them purchasable [71].

What a given tier has actually billed at, tracked release by release: [9]

Table 15. Claude tier price history

Model	Released	Price ($/M in, $/M out)	Change from the prior release in that tier
Claude Sonnet 3.7	Feb 2025	$3 / $15	n/a, first in series
Claude Sonnet 4.6	Feb 2026	$3 / $15	No change in a year
Claude Sonnet 5	Jun 2026	$2 / $10 introductory, then $3 / $15 from 1 Sep 2026	Temporary launch discount only; standard rate unchanged
Claude Opus 4 / 4.1	2025	$15 / $75	n/a, first in series
Claude Opus 4.5 → 4.8	Nov 2025 – May 2026	$5 / $25	One real cut (−67%), delivered by retiring Opus 4 and shipping a new model. Flat across four releases since
Claude Fable 5	Jun 2026	$10 / $50	The newest, most capable tier on the price list, priced *above* Opus 4.8

Over ~16 months and three Sonnet-tier releases, the mid-tier list price hasn't moved: what's changed is capability at a held price, not price at held capability. Opus shows the one genuine price cut, achieved by discontinuing the old model and launching a cheaper one, not by repricing the same product. Fable 5 shows the mechanism can run the other way too: Anthropic's newest frontier tier launched more expensive than the model it sits above.

What this means for the Gartner / Goldman Sachs / Epoch bands used elsewhere as background context [16][17][18]: those studies describe industry-wide compute cost per unit of intelligence falling, which is real, and it does eventually show up as capability moving down the price list (today's Sonnet-level performance reaching Haiku, say). But that's not the same as this deployment's bill falling automatically. Capturing the saving needs an active decision: re-testing a cheaper model against the WP5 evaluation set once it looks good enough, then switching, exactly the Level 2/3 progression already recommended (§7). Waiting passively captures nothing; forced migration on the deprecation schedule above delivers a mix of cheaper, same, or occasionally pricier, depending on what's current when the old tier retires.

Does token volume undercut the price side of the story? Mostly, yes:

- Agentic tasks draw 5–30× the tokens of a standard chat turn [16]. This is directly relevant, since SWISSGEO's own queries are tool-calling/agentic by design (§3, Table 6), not plain chat.
- Live agent deployments typically grow from ~200 to 10,000+ tokens/request within weeks, as history, tool output and system-prompt size accumulate [72].
- This project's own estimate (§4.2–4.3): a complex, multi-step query already runs 2–3× a lean lookup's tokens, before any of the above growth is factored in.

Combined: if tokens/turn rise 3× (a conservative read of the complex-query estimate) while price falls at Goldman Sachs's central ~65%/year, the net effect is 3 × 0.35 ≈ 1.05×: the bill stays roughly flat, not three times lower. At Gartner's more conservative −37%/year, the same 3× token growth pushes the bill up ~90%. Token growth can cancel or reverse a headline price cut; tracking $/token without tracking tokens/turn is optimistic by construction [72].

Practical takeaway. Don't budget against a smooth "prices fall 40–90%/year" curve on today's figure. A migration event roughly every 12–18 months when the recommended model retires and its replacement needs re-validation against WP5 (an operational cost, not just a pricing one); a real chance the per-token price stays flat for a year or more, as Sonnet's has since Feb 2025; and a tokens-per-turn trend more likely to rise than fall as query complexity grows, which can absorb some or all of any headline price cut. The figure worth re-checking quarterly isn't $/million tokens: it's CHF per resolved SWISSGEO query, end to end, since that's the only version of "the bill" that reflects both trends at once.

A note on where the open-weight floor is moving. Gemma 4, this report's "Reference"-tier small open-weight model (§5.4, Table 22), is currently offered by third-party hosts at roughly $0.06–0.99/M input tokens depending on provider and variant, an order of magnitude below even Haiku 4.5 [75]. Current capability is insufficient but the gap is changing.

4.7 The full cost spectrum, in one view

Every scenario and every deployment mode, cheapest to most expensive, with this project's likely position marked. Serverless figures use the realistic, not best-case, optimized numbers from §4.5:

Table 16. Full cost spectrum

Configuration	Monthly cost (CHF, approx.)
Pilot, Haiku 4.5, optimized, serverless	~6
Low (440 turns/day), Sonnet 4.6, lean, optimized (floor case)	~55
Central (715 turns/day), Sonnet 4.6, lean, optimized (floor case)	~90
Central, Sonnet 4.6, complex queries, optimized	**~179–268**
High (1,350 turns/day, swisstopo's own traffic), Sonnet 4.6, lean, optimized (floor case)	~169
Central, Sonnet 4.6, lean, naive	~344
High, Sonnet 4.6, complex queries, optimized	**~337–506**

Configuration	Monthly cost (CHF, approx.)
Central, Mistral Large 3 (La Plateforme, EU vendor), complex queries, naive [74]	**~416–623**
High, Sonnet 4.6, lean, naive	~649
Central, Sonnet 4.6, complex queries, naive	**~687–1,030**
High, Sonnet 4.6, complex queries, naive	**~1,297–1,946**
Self-hosted server, SageMaker, in-country (Zurich), scale-to-zero	~7,900–9,200
Self-hosted server, SageMaker, in-country (Zurich), 24/7, unoptimized	~12,500–21,700
On-premises, small rig, business hours	~11,400–23,300
On-premises, 8×H100-class production rig, full operations team	~39,000–65,000

"Lean" rows are a floor, kept for reference; SWISSGEO's catalogue-search-heavy query pattern (§3, Table 6) makes the complex-query rows the realistic planning basis. On that basis, the planning range spans roughly CHF 179 to CHF 506 per month once optimized, or CHF 687 to CHF 1,946 per month naive, depending on volume. The Mistral Large 3 row is the one cross-vendor comparison in this table.

4.8 Self-hosting

If sovereignty ever requires it, here's the full three-year cost:

Table 17. Self-hosting, three-year cost

Option	Annual, all-in	3-year total
Small rig (2 GPUs, business-hours, 0.5–1 FTE)	CHF 137,000–280,000	CHF 0.4–0.85M
Production rig (8×H100-class, 1.5–2.5 FTE)	CHF 470,000–780,000	CHF 1.4–2.3M
SageMaker, in-country, 24/7, 0.2–0.5 FTE	CHF 150,000–260,000	CHF 0.45–0.8M
Bedrock scale-to-zero (Custom Model Import)	~CHF 95,000	~CHF 0.3M

5. Model options

5.1 Reading the tables below

Two things worth stating before the comparison starts. "On Bedrock" means the model can be called through swisstopo's existing AWS account with no new vendor contract. "Production evidence" means the model has been shown to work for real users in a live deployment, not just scored well on a benchmark.

Where a formal benchmark sits behind a verdict rather than production evidence: geospatial tool-use specifically is scored in GeoBenchX [20]; multi-step MCP tool orchestration in MCPMark [21] and τ²-bench [22]; the failure mode this project cares about most — recovering from an error mid-conversation, or saying "no data exists" instead of guessing (§3) — in 'When Agents Fail to Act' [23] and SciAgentGym [24]; and handling a large catalogue (§3's ~16,000 datasets) in context is the domain of the long-context benchmarks RULER and NoLiMa [25]. None of these are geodata-specific end-to-end proof; they're the closest published evidence for the sub-skills SWISSGEO's assistant actually needs.

5.2 Models in one list

Table 18. Model overview

Model	Vendor	Jurisdiction	Openness	On Bedrock?	EU/CH path	Production evidence	Verdict
Claude Opus 4.8	Anthropic	[US]	Closed	Yes	`eu.` profile available	Quality-ceiling role in Claude deployments incl. the GOV.UK Chat family [5][6]	**Ready**
Claude Sonnet 4.6	Anthropic	[US]	**Closed**	Yes	`eu.` profile available	**GOV.UK Chat runs on this stack, live, national scale; proved for complex use-cases** [5][6]	**Ready, recommended default**
Claude Sonnet 5	Anthropic	[US]	Closed	Yes	Global routing only, no `eu.` profile yet	Newest, ~2 weeks of field data	**Ready** on capability; **Caveat** on EU routing
Claude Haiku 4.5	Anthropic	[US]	Closed	Yes	`eu.` profile available	Matches the prior mid-tier's agent performance at a third of the price	**Ready** (routing tier)
Claude Fable 5	Anthropic	[US]	Closed	Yes	`eu.` profile available, but requires a provider-data-sharing opt-in and forces temperature=1.0	Newest, most powerful tier; overkill for this use case	**Caveat**
GPT-5.5 / 5.4 + Codex	OpenAI	[US]	Closed	Yes, US regions only	Direct API offers EU residency incl. Switzerland	Strong agentic claims; some vendor benchmark comparisons used unequal settings	**Caveat**

Model	Vendor	Jurisdiction	Openness	On Bedrock?	EU/CH path	Production evidence	Verdict
GPT-5.6 (Sol/Terra/ Luna)	OpenAI	[US]	Closed	No	Direct API + EU residency	Newest tier	**Reference** (watch item)
Gemini 3.1 Pro / 3.5 Flash	Google	[US]	Closed	No, not on AWS at all	Vertex EU multi-region; no Zurich pinning	Strongest published multilingual scores [33]	**Reference** (wrong cloud for this stack)
Grok 4.3	xAI	[US]	Closed	Yes, US regions only	None documented	Signed only the safety chapter of the EU AI-Act code of practice [44]	**Not viable** here
Amazon Nova 2 Lite	Amazon	[US]	Closed	Yes, native	EU regions yes, not invocable from Zurich	Evidence is largely vendor-reported [56]	**Caveat** (executor tier only)
Cohere Command A+	Cohere / Aleph Alpha	[US/EU, merged]	Open-weight	No, not on Bedrock in any form	Self-host only	Citation-native RAG design	**Reference**
Mistral Large 3	Mistral AI	[EU] (France)	Open-weight (Apache 2.0)	Yes, US/ APAC regions only	Self-host, or Mistral's own EU platform, for EU residency	1M+ French civil servants; French armed-forces framework tests [80]	**Caveat** (the strongest European option, but markedly behind Claude on demonstrated agentic capability, not a peer: no live tool-heavy agent deployment exists, and its function calling has a documented tool-description over-weighting failure mode)
Ministral 3 / Devstral 2 / Magistral / Voxtral	Mistral AI	[EU]	Open-weight	Yes, EU regions	`eu.`-region native	Smaller-task / router role	**Ready** (router tier)
gpt-oss-120b / 20b	OpenAI	[US]	Open-weight (Apache 2.0)	Yes	EU regions	AI Sweden (on-prem), Orange, Snowflake Cortex	**Caveat** (known failure mode on long tool chains)
Gemma 4	Google	[US]	Open-weight (Apache 2.0)	No	Self-host	Strong small model	**Reference**
IBM Granite 4.x	IBM	[US]	Open-weight (Apache 2.0)	No, watsonx not Bedrock	Self-host	"Western Qwen" positioning	**Reference**
GLM-5.x	Zhipu AI	[CN]	Open-weight	Yes, some regions	Self-host for full control	Coinbase adopted; scores at frontier level on agentic benchmarks	**Not viable** as core (eval reference only)

Model	Vendor	Jurisdiction	Openness	On Bedrock?	EU/CH path	Production evidence	Verdict
DeepSeek V3.2 / V4	DeepSeek	[CN]	Open-weight	Yes, some EU regions	Self-host for full control	Documented tool-call parser instability	**Not viable** as core (eval reference only)
Qwen3.x	Alibaba	[CN]	Open-weight (Apache 2.0)	Yes, some EU regions	Self-host for full control	Very large adoption globally; parser friction documented	**Caveat** (self-hosted)
Kimi K2.x	Moonshot AI	[CN]	Open-weight (MIT)	Yes	Self-host for full control	Ties GPT-5.5 on a coding benchmark	**Not viable** as core (eval reference only)
Apertus 8B/70B	Swiss AI Initiative (ETH Zürich, EPFL, CSCS)	[CH]	**Open-source** (weights + data + code, Apache 2.0)	No	SageMaker (Zurich), Swisscom, Infomaniak	CHUV, Canton Ticino translation; zero agentic production deployments found	**Caveat** as core; **Ready** as sovereign sidecar, watched rather than tested
OLMo 3 / 3.1	Allen Institute for AI	[US]	Open-source	No	Self-host	Zero named production deployments	**Not viable** yet (methodological reference only)
EuroLLM	EU-funded consortium	[EU]	Open-source	No	Self-host	No native tool calling	**Reference** (multilingual auxiliary only)
Llama 4	Meta	[US]	Restricted licence	Yes	None	Benchmark controversy; licence prohibits EU-domiciled entities from its multimodal models	**Not viable**

5.3 The same models, grouped by family

Proprietary frontier models

Table 19. Proprietary frontier models

Model	Jurisdiction	Bedrock + EU from Switzerland?	Verdict
Claude (Opus 4.8 / Sonnet 4.6 / Haiku 4.5)	[US]	Yes	**Ready, recommended**
GPT-5.5/5.4	[US]	Bedrock yes, EU routing no (US regions only)	**Caveat**
Gemini 3.x	[US]	Not on Bedrock at all	**Reference** (wrong cloud)
Grok 4.3	[US]	Bedrock yes, EU routing no	**Not viable** here
Nova 2 Lite	[US]	Bedrock yes, not invocable from Switzerland	**Caveat** (executor tier only)
Command A+	[US/EU]	Not on Bedrock	**Reference**

Open-weight models

Table 20. Open-weight models

Model	Jurisdiction	Verdict
Mistral Large 3	[EU]	**Caveat** (European anchor, a real capability step down from Claude; EU residency comes from Mistral's own La Plateforme platform, not Bedrock, where Mistral is US/APAC-only; self-hosting is an option)
gpt-oss-120b	[US]	**Caveat** (long-tool-chain formatting issue)
Gemma 4 / Granite 4	[US]	**Reference**
GLM / DeepSeek / Qwen / Kimi	[CN]	See the jurisdiction split below

Swiss-specific: Apertus, the one model here actually built, trained and hostable entirely in Switzerland, and fully open-source rather than merely open-weight.

Table 21. Swiss-specific: Apertus

Model	Jurisdiction	Openness	Verdict
Apertus 8B/70B	[CH]	Open-source (the strongest transparency)	**Caveat** as agent core; **Ready** as the sovereign sidecar

5.4 Open-source and open-weight models, in depth

Table 22. Open-source and open-weight models, in depth

Model	Licence	What's released	Trained by	Runs with zero vendor contact?	Verdict
Mistral Large 3 / Ministral family	Apache 2.0 [32]	**Weights only**	Mistral AI SAS, Paris, France [EU] [62]	Yes	**Caveat** (the European anchor)
gpt-oss-120b / 20b	Apache 2.0 [68]	Weights only	OpenAI Group PBC, USA [US]	Yes	**Caveat**
Gemma 4	Apache 2.0 (the first Apache-licensed Gemma release) [47]	Weights only	Google (Alphabet Inc.), USA [US]	Yes	**Reference**
IBM Granite 4.x	Apache 2.0 [67]	Weights only	IBM Corporation, USA [US]	Yes	**Reference**
GLM-5.x	MIT [66]	Weights only	Zhipu AI, Beijing, China [CN]	Yes, technically (see the jurisdiction table below)	**Not viable** as core
DeepSeek V3.2 / V4	MIT [66]	Weights only	DeepSeek, Hangzhou, China [CN]	Yes, technically (see the jurisdiction table below)	**Not viable** as core
Qwen3.x	Apache 2.0 [66]	Weights only	Alibaba Group, Hangzhou, China [CN]	Yes, technically (see the jurisdiction table below)	**Caveat** (self-hosted)

Model	Licence	What's released	Trained by	Runs with zero vendor contact?	Verdict
Kimi K2.x	Modified MIT: standard MIT below 100M monthly active users or $20M in monthly revenue; above that threshold, the product must display "Kimi K2" in its interface [64]	Weights only	Moonshot AI, Beijing, China [CN]	Yes, technically (see the jurisdiction table below)	**Not viable** as core
Apertus 8B/ 70B	Apache 2.0 [26]	**Weights, training data and code** (the most complete release in this table)	Swiss AI Initiative (ETH Zürich, EPFL, CSCS), Switzerland [CH]	Yes	**Caveat** as core; **Ready** as sidecar
OLMo 3 / 3.1	Apache 2.0 [46]	Weights, training data and code	Allen Institute for AI, Seattle, USA [US] (non-profit research institute, not a commercial vendor)	Yes	**Not viable** yet
EuroLLM	Apache 2.0 [54]	Weights, training data and code (near-complete)	EU-funded research consortium, institutions across several member states [EU]	Yes	**Reference**
Llama 4	Meta's own Llama 4 Community Licence, not Apache, not MIT, and not an OSI-approved open-source licence [65]	Weights only, with usage restrictions attached	Meta Platforms, Inc., USA [US]	Technically self-hostable, but the licence itself bars EU-domiciled individuals and companies from its multimodal models, regardless of hosting arrangement [65]	**Not viable**

Three precisions, since "open" gets used loosely elsewhere. Apache 2.0 and MIT are genuinely permissive, OSI-approved: run indefinitely, no vendor contact, no retroactive licence changes. "Open-weight" (most rows above) is narrower than "open-source": the trained model is released, but not the data or code used to train it, which matters for provenance and eCH-0272's transparency expectations, not for self-hosting. And Llama 4 is the one model here marketed as "open" that isn't: its bespoke Meta licence with a geographic carve-out is a licensing choice, not a technical or jurisdictional one, hence its own category in this table.

5.5 Chinese open-weight models

Table 23. Chinese open-weight models: API vs. self-hosted

Model	Via the vendor's own API	Self-hosted (weights downloaded, run on AWS or own infrastructure)
GLM-5.x, DeepSeek V3.2/V4, Qwen3.x, Kimi K2.x	**Not viable** (data flows to PRC-jurisdiction servers, and Switzerland has no data-adequacy finding for China)	**Caveat** — the vendor has no access to the data or the running system

Note: bias and weaker safety-robustness remain baked into the weights themselves, potentially mitigated by guardrails and evaluation.

5.6 Apertus, in more detail

Table 24. **Apertus, in more detail**

Dimension	Today	At the promised v1.5 (undated)
Tool calling (the core capability an agent needs)	Not supported in the standard way; only manual workarounds exist	Promised, no release date announced
Instruction following	Measurably behind a same-size US budget model in 8 months of live production data (Liip/ZüriCityGPT)	Expected to improve; unverified
Romansh	The best-scoring published model in independent evaluations	Same strength should carry forward
Swiss German	Comprehension is reasonable; generated text was judged "practically unusable" in independent testing	Unknown
Where it can run	SageMaker (including Zurich), Swisscom, Infomaniak, PublicAI	Unchanged
Real-world adoption so far	Canton Ticino (translation), CHUV (clinical trials); zero agentic deployments found anywhere	Worth keeping an eye on once tool calling ships

Role in this project today: the Romansh/Swiss-German layer alongside the main assistant, where it already performs well. Worth keeping an eye on it, if and when v1.5's tool calling matures.

5.7 Production-readiness verdict, condensed

Table 25. **Production-readiness verdict, condensed**

Candidate	Verdict	The evidence in one line
Claude Sonnet 4.6/5 + Haiku 4.5 (Bedrock)	**Ready** (production agent core)	GOV.UK Chat live at national scale on this exact stack [5][6]
Mistral Large 3 + Ministral	**Caveat, clearly behind Claude**	Large public-sector footprint, but assistant/RAG-style, not tool-heavy agentic [80]
Amazon Nova 2 Lite	**Caveat** (executor tier only)	Real deployments exist, but evidence is largely vendor-reported; not reachable from Zurich
gpt-oss-120b	**Caveat**	Known formatting failure after long chains of tool calls, exactly this project's search-then-identify pattern
Qwen3.x (self-hosted)	**Caveat**	Very large adoption elsewhere; documented serving-software friction we'd own
Apertus 70B	**Caveat** as core; **Ready** as sidecar	No tool calling until v1.5; strong as a Romansh/dialect layer today
GLM-5.x / DeepSeek	**Not viable** (eval reference only)	Genuinely frontier-capable, but blocked by procurement and reputational concerns for a federal deployment
Llama 4	**Not viable**	Frozen product line; licence actively hostile to EU-domiciled use
OLMo 3	**Not viable** yet	Zero production deployments found anywhere; a transparency reference, not a deployment candidate

6. Platform and residency options

"Sovereignty" is here as a scale with five steps. The same software runs on every step, and moving between them is a deployment change.

6.1 The sovereignty scale

Table 26. The sovereignty scale

Step	What it is	Jurisdiction	Who controls it	When it's the right step
1. Bedrock `eu.` profiles	Requests guaranteed to process inside EU data centres	[EU] (AWS/ Anthropic-controlled)	Vendor	**Today's default** (sufficient for the stated "möglichst EU/ national" preference)
2. Open-weight models on Bedrock (EU regions)	Same serverless model, swapped to an open-weight one	[EU/US/CN depending on model]	Vendor hosts, licence governs use	Cost-sensitive tiering, or evaluation of alternatives
3. SageMaker in Zurich	Self-hosted, in-country, any open model including Apertus	[CH]	swisstopo/askEarth operates the instance; AWS controls the region	A firm mandate for in-country processing
4. Swiss providers (Swisscom, Infomaniak, PublicAI)	Vendor-hosted, physically in Switzerland	[CH]	Swiss vendor	A vendor-independent Swiss fallback
5. On-premises	Fully in-house hardware	[CH], complete control	swisstopo/askEarth, entirely	Only if physical, air-gapped control is mandated by policy; priced honestly in §4

6.2 Jurisdiction, properly explained: who owns what, where, and what that actually means

The [CH]/[EU]/[US]/[CN] tag used throughout this report is a simplification, and it's worth unpacking once, properly, rather than leaving it as a one-letter label. "Jurisdiction" actually bundles three separate questions that can each point to a different country:

1. **Who legally owns and operates the vendor**: its country of incorporation, which determines whose courts and laws can compel it to act.
2. **Where the data is technically processed**: which physical region's servers handle a given request, relevant mainly to local seizure or access rules.
3. **Where the model was trained**: relevant to provenance and to standards like eCH-0272, but not to who can compel access to a live conversation.

The tag tracks the first of these, since it matters most for compelled-disclosure risk. Clearest example: the US CLOUD Act (2018) lets US law enforcement compel a US-incorporated company to hand over data it controls, wherever it physically sits [63]. Bedrock's EU-resident routing keeps processing inside EU data centres, satisfying the SGS report's residency preference, but Anthropic remains a US public benefit corporation, so the vendor-level dependency the SGS report's Microsoft quote warns about is real regardless of region. This is why §6.1's sovereignty scale treats weight-portability, not the region flag, as the durable guarantee.

Table 27. Vendor jurisdiction

Vendor / model family	Legal entity	Headquarters	Whose law can compel disclosure	What this means here
Anthropic (Claude)	Anthropic PBC, a Delaware public benefit corporation [59]	San Francisco, USA	US law, incl. the CLOUD Act	The recommended stack. EU-resident routing controls *where processing happens*, not Anthropic's status as a US company
Amazon (Bedrock platform; Nova models)	Amazon.com, Inc., Delaware	Seattle, USA	US law	Bedrock is US-owned infrastructure wherever its servers physically sit; the same reasoning applies to the hosting platform itself, not just the models on it
OpenAI (GPT family)	OpenAI Group PBC, a Delaware public benefit corporation, controlled by the non-profit OpenAI Foundation since the October 2025 restructuring [60]	San Francisco, USA	US law	Same category as Anthropic; currently without Bedrock-EU availability (§5.3, Table 19)
Google (Gemini)	Google LLC, a subsidiary of Alphabet Inc., Delaware	Mountain View, USA	US law	Not reachable via Bedrock at all (§5.3, Table 19); the jurisdiction question is moot here for this project
xAI (Grok)	X.AI Corp., incorporated in Nevada, March 2023 [61]	San Francisco, USA	US law	US-regions-only on Bedrock; not viable here on evidence grounds regardless (§5.3, Table 19)
Cohere (Command A+)	Cohere Inc., incorporated in Canada	Toronto, Canada	Canadian law (a genuinely different jurisdiction from the US majority above)	Not on Bedrock in any form; a reference point only
Mistral AI (Large 3, Ministral)	Mistral AI SAS [62]	Paris, France	French and EU law	The one frontier-adjacent vendor in this comparison actually incorporated inside the EU
Meta (Llama 4)	Meta Platforms, Inc., Delaware	Menlo Park, USA	US law	Also carries its own licence-level exclusion of EU-domiciled users for the multimodal models, a contractual restriction on top of, not instead of, the jurisdiction question
Allen Institute for AI (OLMo)	Non-profit research institute	Seattle, USA	US law, but there's no commercial API to compel access through; the weights are simply published	Lowers the practical relevance of jurisdiction for this specific model
Zhipu AI (GLM)	Zhipu AI, incorporated in China	Beijing, China	Chinese law, including national intelligence and cybersecurity legislation that can compel a China-incorporated company to cooperate with state requests	Applies to Zhipu's own hosted API. Does not apply in the same way to a copy of the published weights running with no connection back to Zhipu (§5.5's Chinese-model table)
DeepSeek	DeepSeek, incorporated in China	Hangzhou, China	Chinese law	Same reasoning as Zhipu
Alibaba (Qwen)	Alibaba Group	Hangzhou, China	Chinese law	Same reasoning as Zhipu

Vendor / model family	Legal entity	Headquarters	Whose law can compel disclosure	What this means here
Moonshot AI (Kimi)	Moonshot AI, incorporated in China	Beijing, China	Chinese law	Same reasoning as Zhipu
IBM (Granite)	International Business Machines Corporation	Armonk, USA	US law	Offered mainly through IBM's own watsonx platform, not Bedrock
Swiss AI Initiative (Apertus)	A public academic consortium (ETH Zürich, EPFL, and the Swiss National Supercomputing Centre), not a commercial company at all	Zürich / Lausanne / Lugano, Switzerland	Swiss law; as public institutions, not exposed to a foreign compelled-disclosure regime the way a commercial vendor is	The only entry in this report where the owner is Swiss public institutions rather than any company, the clearest sovereignty story available, alongside the capability gaps documented in §5.6 (Table 24)
EuroLLM	An EU-funded, multi-institution research consortium	Institutions across several EU member states	EU law, distributed across the member states involved	A public research project, like Apertus, not a commercial vendor

This also answers the SGS report's sharpest concern: a US hyperscaler's regional promises don't remove the underlying jurisdiction, and that applies to AWS as much as Microsoft. For public, non-personal geodata, the honest framing is that the residual risk is vendor dependency, not data confidentiality. The scale's guarantee against that dependency is weights that can be taken anywhere under Apache/MIT licensing, not the region flag on a server.

6.3 Bedrock vs. other clouds and direct vendor APIs

If a frontier model that AWS can't serve from an EU region ever becomes the deciding factor, here's the menu. All figures are naive (no prompt caching, no routing) on a single, shared complex-query definition (2–3× the base token count) for cross-vendor comparability, not the full naive/complex spectrum from §4:

Table 28. Bedrock vs. other clouds and direct vendor APIs

Provider	Model & platform	Deployed on	Jurisdiction	CHF/turn (naive, complex query)	CHF/month (Central, 715 turns/day, naive)	Residency reality
Google	Gemini 3.5 Flash, Vertex AI (EU multi-region) [76]	Google Cloud	[EU boundary / US vendor]	0.0104–0.0156	~CHF 227–340	EU boundary, no Swiss region
Mistral AI	Mistral Large 3, La Plateforme [74]	Mistral's own infrastructure	[EU]	0.0112–0.0168	~CHF 243–364	EU datacentres, EU vendor
OpenAI	GPT-5.6 "Terra" tier, direct API + EU residency [77]	OpenAI's own infrastructure	[EEA incl. CH / US vendor]	0.0174–0.0261	~CHF 378–567	Zero data retention
Anthropic	Claude Sonnet 4.6, Amazon Bedrock `eu.` (status quo) [9]	AWS (Amazon Bedrock)	[EU / US vendor]	0.0182–0.0273	~CHF 396–594	EU regions, existing AWS setup (**recommended**)
OpenAI (via Microsoft)	GPT-5.5, Azure OpenAI, Switzerland North, Global Standard [78]	Microsoft Azure	[CH data-at-rest / US vendor, global inference]	0.0316–0.0474	~CHF 687–1,030	Data stored in Switzerland, but processed anywhere in Microsoft's global fleet, a materially weaker guarantee than it sounds

Note: all prices in USD, converted at the same ≈0.81 CHF/USD rate throughout [73]. This table's query basis is shorter than §4's own (see §4.3 for this project's real naive/lean and naive/complex figures); the point here is relative ordering between vendors, not absolute cost.

6.4 What's actually available from Switzerland

as of 14.07.2026

Table 29. What's actually available from Switzerland

Resource	Available in Zurich (`eu-central-2`)?	Where it actually runs otherwise
Claude 3 Haiku, Titan Text Embeddings V2	Yes, true in-region	Zurich
Current Claude generation (Haiku 4.5, Sonnet 4.5/4.6, Opus 4.5–4.8, Fable 5)	Via `eu.` cross-region routing	EU destination regions, including Zurich for several of them
Claude Sonnet 5	No `eu.` profile yet	Global routing
High-end GPU instances (H100/L40S-class)	Not available	Nearest: Frankfurt
Entry-level GPU instances (L4, 24GB)	Available	Zurich
Bedrock Custom Model Import (scale-to-zero self-hosting)	Not available	Frankfurt and a handful of US regions
Open-weight serverless models (DeepSeek, Qwen, Kimi, gpt-oss, GLM Flash, Ministral)	None invocable from Zurich	Ireland, Milan, London, Frankfurt, Stockholm
Mistral Large 3 (serverless)	Not in any EU region	US/APAC only

No committed date exists for Zurich reaching this parity [79].

7. Recommendation and plan

7.1 Rollout levels

Table 30. Rollout levels

Level	Action	Gate to move forward	Expected effect
1: Deploy	Claude Sonnet 4.6 via Bedrock `eu.`, built on Bedrock AgentCore (the current agent framework)	None	Live pilot
2: Optimize	Potential caching, routing	The WP5 evaluation shows quality, cost	Cost reduction
3: Watch	No active testing of alternatives; Mistral Large 3, Apertus and other models stay monitored	If a concrete trigger arises	-

8. Risks

8.1 Risk register

Table 31. Risk register

Risk	Likelihood / impact	Mitigation
Real adoption exceeds the grounded scenarios in §4.2 (Table 9)	Low / Low	Cost scales linearly and stays small even at High (~1,350 turns/day, already scaled to swisstopo's own traffic)
A vendor changes a model or its price	High / Low	Model identifier kept as a simple configuration value; observe
No EU-resident routing yet for the newest models (e.g. Sonnet 5)	Medium / Low	Sonnet 4.6 is the EU-resident default; this pattern (a delay, then an `eu.` profile follows) has repeated with every recent Claude release
Dependency on a single vendor (AWS/ Anthropic)	Medium / Medium	Develop exit option

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